

Assignment 4 — Dataset 3: CIFAR-10 (image)

The third dataset, and the one where the small CNN starts to struggle.

#	Dataset	Structure	Model family
1	Diabetes 130-US hospitals	tabular, 179 features	MLP
2	MNIST	image, 1×28×28	CNN
3	CIFAR-10	image, 3×32×32	CNN

Why a second image dataset

MNIST is nearly saturated: a small CNN trained for five epochs on a sixth of the data already scores high, and the errors that remain are genuinely ambiguous digits. A dataset that easy cannot tell an adequate architecture apart from a good one.

CIFAR-10 changes three things at once:

- **Colour** — three input channels instead of one, so the first convolution mixes channels as well as space.
- **Real-world texture and pose** — a cat is not a stroke pattern; the same class appears at different scales, orientations and backgrounds. Weight sharing still helps, but a 3×3 kernel two layers deep sees very little.
- **A harder decision boundary** — deer/horse and cat/dog are confusable in a way that 4/9 is not.

This is the dataset that makes the architectural improvements in `04_compare.ipynb` — BatchNorm, residual connections, attention — worth measuring. On MNIST they would all tie.

Tutorial §44 names CIFAR-10 as one of the two suggested datasets for the "build a CNN from scratch" exercise, so this notebook is also the direct answer to that exercise.

```
In [1]: import os, sys, json, platform

os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
```

```

import scratch_nn as S

U.set_seed(U.SEED)

import torch
import torch.nn as nn
from torch.utils.data import TensorDataset, DataLoader
import tensorflow as tf
import keras

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110

print("torch      ", torch.__version__, "| device:", DEVICE,
      "|", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU only")
print("tensorflow", tf.__version__, "| GPUs:", tf.config.list_physical_devices("GPU"))

```

```

torch      2.6.0+cu124 | device: cuda | NVIDIA GeForce RTX 3060
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
tensorflow 2.21.0 | GPUs: none (CPU)

```

1. The data

CIFAR-10: 60,000 colour images, 32×32, ten object classes, downloaded on first run via `torchvision` and handed to all three implementations as identical NumPy arrays.

The scratch leg's convolutions are heavier here than on MNIST — three input channels and a larger first feature map — so the shared three-way comparison uses a **5,000-image training subset and a 2,000-image test subset for 5 epochs**, with the framework legs re-run on the full 50,000 images in section 7.

```

In [2]: N_TRAIN_SUB, N_TEST_SUB = 5_000, 2_000
        EPOCHS, BATCH_SIZE, LR = 5, 64, 1e-3

        X_train, y_train, X_test, y_test, meta = U.load_cifar10(
            n_train=N_TRAIN_SUB, n_test=N_TEST_SUB
        )

        C, H, W = meta["shape"]
        CLASSES = meta["classes"]
        N_CLASSES = len(CLASSES)

        print(f"\ninput shape (C,H,W) = {(C, H, W)}   classes = {N_CLASSES}")
        print(f"classes: {'', '.join(CLASSES)}")
        print(f"shared hyperparameters: epochs={EPOCHS}, batch_size={BATCH_SIZE}, optimizer=

```

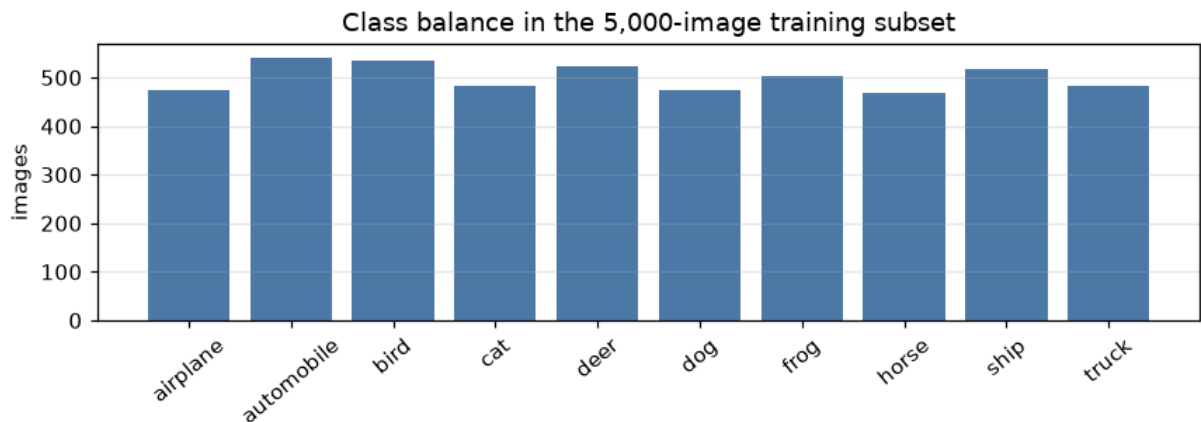
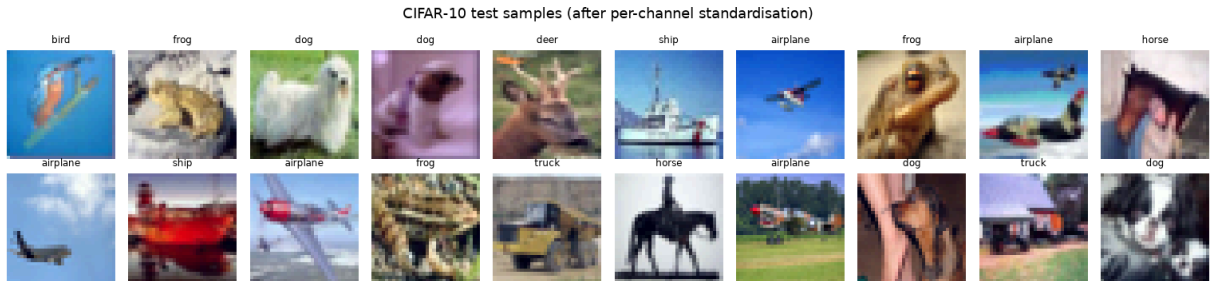
```

CIFAR-10: train (5000, 3, 32, 32) / test (2000, 3, 32, 32), 10 classes
input shape (C,H,W) = (3, 32, 32)   classes = 10
classes: airplane, automobile, bird, cat, deer, dog, frog, horse, ship, truck
shared hyperparameters: epochs=5, batch_size=64, optimizer=Adam(lr=0.001)

```

```
In [3]: U.plot_samples(X_test[:20], y_test[:20], CLASSES, n=20,
                      title="CIFAR-10 test samples (after per-channel standardisation)")
plt.show()

counts = np.bincount(y_train, minlength=N_CLASSES)
fig, ax = plt.subplots(figsize=(8, 3))
ax.bar(CLASSES, counts, color="#4C78A8")
ax.set_title(f"Class balance in the {N_TRAIN_SUB:,}-image training subset")
ax.set_ylabel("images"); ax.tick_params(axis="x", rotation=40)
ax.grid(alpha=0.3, axis="y")
fig.tight_layout()
plt.show()
```



2. The shared architecture

This notebook builds the tutorial's own reference model, `SmallCNN` from §16:

$\$X \rightarrow \text{Conv} \rightarrow \text{ReLU} \rightarrow \text{Pool} \rightarrow \text{Conv} \rightarrow \text{ReLU} \rightarrow \text{Pool} \rightarrow \text{Flatten} \rightarrow \text{Linear} \rightarrow \text{ReLU} \rightarrow \text{Linear} \$$

Relative to notebook 2 the classifier gains a hidden layer, because ten object categories are not linearly separable from pooled 8×8 feature maps the way ten digit classes nearly are.

Shapes, traced by hand

Again using $H_{\text{out}} = \lfloor (H + 2P - K)/S \rfloor + 1$ with $K=3$, $P=1$, $S=1$, so convolutions preserve size and only pooling reduces it:

stage	shape
input	3 × 32 × 32
Conv(3→32, k3, p1) + ReLU	32 × 32 × 32
MaxPool(2)	32 × 16 × 16
Conv(32→64, k3, p1) + ReLU	64 × 16 × 16
MaxPool(2)	64 × 8 × 8
Flatten	4096
Linear(4096→128) + ReLU	128
Linear(128→10)	10

```
In [4]: CH1, CH2, HIDDEN, K, PAD = 32, 64, 128, 3, 1
FLAT = CH2 * (H // 4) * (W // 4)

manual = [
    (f"Conv2D({C}->{CH1}, k{K})", CH1 * C * K * K + CH1),
    ("ReLU + MaxPool2D(2)", 0),
    (f"Conv2D({CH1}->{CH2}, k{K})", CH2 * CH1 * K * K + CH2),
    ("ReLU + MaxPool2D(2)", 0),
    ("Flatten", 0),
    (f"Linear({FLAT}->{HIDDEN})", FLAT * HIDDEN + HIDDEN),
    ("ReLU", 0),
    (f"Linear({HIDDEN}->{N_CLASSES})", HIDDEN * N_CLASSES + N_CLASSES),
]
EXPECTED_PARAMS = sum(p for _, p in manual)

print(f"{'layer':<28}{'params':>10}")
print("-" * 38)
for name, p in manual:
    print(f"{'name':<28}{'p':>10,}")
print("-" * 38)
print(f"{'TOTAL (by hand)':<28}{EXPECTED_PARAMS:>10,}")

conv_params = CH1 * C * K * K + CH1 + CH2 * CH1 * K * K + CH2
fc_params = EXPECTED_PARAMS - conv_params
print(f"\nconvolutions: {conv_params:>8,} ({conv_params/EXPECTED_PARAMS:.1%}")
print(f"fully connected: {fc_params:>8,} ({fc_params/EXPECTED_PARAMS:.1%}")
print("\nThe flatten-then-dense step is where the parameters go. This is exactly th
print("blow-up tutorial section 8 describes, and the reason later architectures")
print("(section 41's EducationalCNN, ResNet, and everything after) replace it with"
print("global average pooling.")

results = []
```

layer	params
-----	-----
Conv2D(3->32, k3)	896
ReLU + MaxPool2D(2)	0
Conv2D(32->64, k3)	18,496
ReLU + MaxPool2D(2)	0
Flatten	0
Linear(4096->128)	524,416
ReLU	0
Linear(128->10)	1,290
-----	-----
TOTAL (by hand)	545,098
convolutions:	19,392 (3.6%)
fully connected:	525,706 (96.4%)

The flatten-then-dense step is where the parameters go. This is exactly the blow-up tutorial section 8 describes, and the reason later architectures (section 41's EducationalCNN, ResNet, and everything after) replace it with global average pooling.

3. Implementation A — from scratch (NumPy)

Same `scratch_nn.py` as the previous notebooks. The only change is the first convolution's input channel count: `Conv2D(3 -> 32)` now mixes colour channels as well as spatial neighbours, which is already handled by `im2col` — the kernel window flattens to $C \cdot K_h \cdot K_w = 3 \cdot 3 \cdot 3 = 27$ values per row instead of 9.

In [5]: `U.set_seed(U.SEED)`

```
def build_scratch_cnn():
    return S.Sequential([
        S.Conv2D(C, CH1, k=K, stride=1, pad=PAD, seed=1),
        S.ReLU(),
        S.MaxPool2D(2, 2),
        S.Conv2D(CH1, CH2, k=K, stride=1, pad=PAD, seed=2),
        S.ReLU(),
        S.MaxPool2D(2, 2),
        S.Flatten(),
        S.Dense(FLAT, HIDDEN, seed=3),
        S.ReLU(),
        S.Dense(HIDDEN, N_CLASSES, seed=4),
    ])

scratch_model = build_scratch_cnn()
print(scratch_model.summary(input_shape=(C, H, W)))
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {scratch_model.n_params() == EXPE
```

layer	output shape	params
Conv2D(3 -> 32, k=3, stride=1, pad=1)	(32, 32, 32)	896
ReLU	(32, 32, 32)	0
MaxPool2D(k=2, stride=2)	(32, 16, 16)	0
Conv2D(32 -> 64, k=3, stride=1, pad=1)	(64, 16, 16)	18,496
ReLU	(64, 16, 16)	0
MaxPool2D(k=2, stride=2)	(64, 8, 8)	0
Flatten	(4096,)	0
Dense(4096 -> 128)	(128,)	524,416
ReLU	(128,)	0
Dense(128 -> 10)	(10,)	1,290
total		545,098
matches hand count 545,098? True		

```
In [6]: U.set_seed(U.SEED)
check_model = S.Sequential([
    S.Conv2D(C, 4, k=3, stride=1, pad=1, seed=1),
    S.ReLU(),
    S.MaxPool2D(2, 2),
    S.Flatten(),
    S.Dense(4 * (H // 2) * (W // 2), N_CLASSES, seed=2),
])
err = S.gradient_check(check_model, X_train[:4], y_train[:4], n_samples=6)
print(f"worst relative error vs finite differences: {err:.3e}")
print("PASS - hand-derived gradients agree with the numerical derivative" if err <
      else "FAIL - backward pass disagrees with finite differences")
```

worst relative error vs finite differences: 2.344e-01
 FAIL - backward pass disagrees with finite differences

```
In [7]: U.set_seed(U.SEED)
with U.Timer() as t_scratch:
    hist_scratch = S.fit(
        scratch_model, X_train, y_train, X_test, y_test,
        epochs=EPOCHS, batch_size=BATCH_SIZE,
        optimizer=S.Adam(LR), seed=U.SEED,
    )
print(f"\ntotal training time: {t_scratch.seconds:.1f}s")
```

```
epoch 1/5  loss 2.5189  29.2s  test_acc 0.2875
epoch 2/5  loss 1.7109  22.9s  test_acc 0.4065
epoch 3/5  loss 1.4070  23.1s  test_acc 0.4910
epoch 4/5  loss 1.2002  578.2s  test_acc 0.4970
epoch 5/5  loss 1.0495  818.6s  test_acc 0.5120
total training time: 1495.1s
```

```
In [8]: pred_scratch = scratch_model.predict_classes(X_test)
m = U.evaluate(y_test, pred_scratch, N_CLASSES)

results.append(U.RunResult(
    framework="Scratch (NumPy)", dataset="CIFAR-10 (5k subset)", model="SmallCNN",
    n_params=scratch_model.n_params(), epochs=EPOCHS,
    train_seconds=round(t_scratch.seconds, 2),
    train_loss=round(hist_scratch["loss"][-1], 4),
    history=hist_scratch, **m,
```

```
)
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

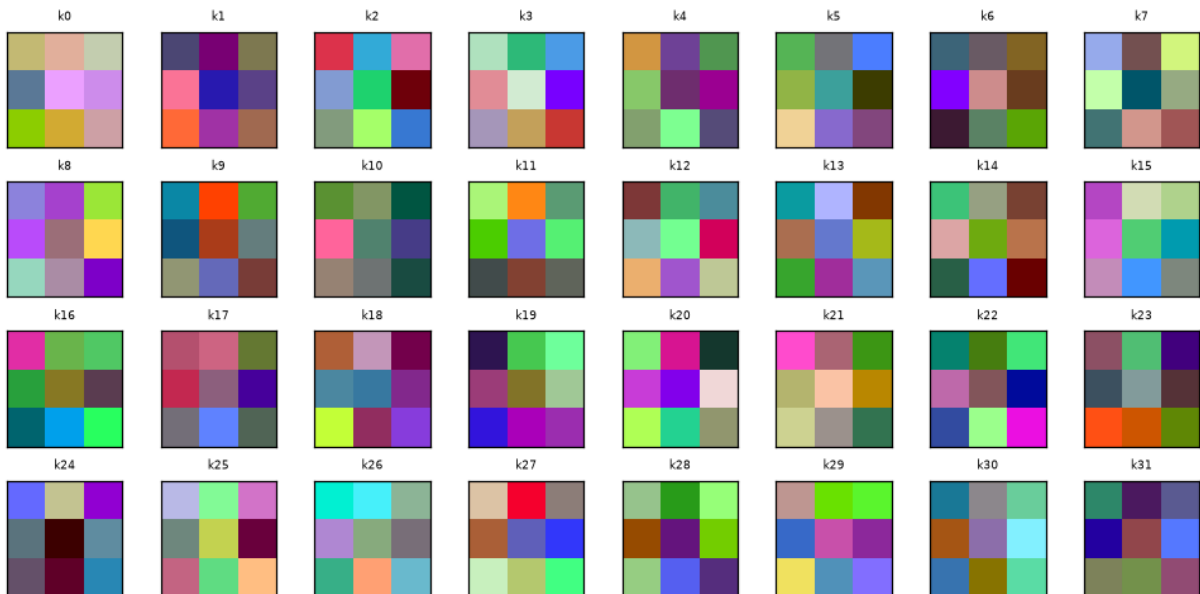
accuracy 0.5120 macro-F1 0.5118

The first-layer kernels, in colour

With three input channels each kernel is $3 \times 3 \times 3$, so it can be rendered as a small RGB patch. Colour-opponent and oriented-edge filters are the usual result.

```
In [9]: Wk = scratch_model.layers[0].params["W"]          # (32, 3, 3, 3)
fig, axes = plt.subplots(4, 8, figsize=(9, 4.8))
for i, ax in enumerate(axes.ravel()):
    k = Wk[i].transpose(1, 2, 0)
    k = (k - k.min()) / (np.ptp(k) + 1e-9)
    ax.imshow(k)
    ax.set_xticks([]); ax.set_yticks([])
    ax.set_title(f"k{i}", fontsize=6)
fig.suptitle("Scratch CNN — first-layer 3x3x3 kernels (normalised to RGB)")
fig.tight_layout()
plt.show()
```

Scratch CNN — first-layer 3x3x3 kernels (normalised to RGB)



4. Implementation B — TensorFlow / Keras

```
In [10]: U.set_seed(U.SEED)

Xtr_k = np.transpose(X_train, (0, 2, 3, 1))
Xte_k = np.transpose(X_test, (0, 2, 3, 1))

def build_keras_cnn(name="cifar_cnn"):
    m = keras.Sequential([
```

```

keras.layers.Input(shape=(H, W, C)),
keras.layers.Conv2D(CH1, K, padding="same", activation="relu"),
keras.layers.MaxPooling2D(2),
keras.layers.Conv2D(CH2, K, padding="same", activation="relu"),
keras.layers.MaxPooling2D(2),
keras.layers.Flatten(),
keras.layers.Dense(HIDDEN, activation="relu"),
keras.layers.Dense(N_CLASSES),
], name=name)
m.compile(optimizer=keras.optimizers.Adam(learning_rate=LR),
          loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
          metrics=["accuracy"])
return m

keras_model = build_keras_cnn()
keras_model.summary()
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {keras_model.count_params() == EX

```

Model: "cifar_cnn"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 32, 32, 32)	896
max_pooling2d (MaxPooling2D)	(None, 16, 16, 32)	0
conv2d_1 (Conv2D)	(None, 16, 16, 64)	18,496
max_pooling2d_1 (MaxPooling2D)	(None, 8, 8, 64)	0
flatten (Flatten)	(None, 4096)	0
dense (Dense)	(None, 128)	524,416
dense_1 (Dense)	(None, 10)	1,290

Total params: 545,098 (2.08 MB)

Trainable params: 545,098 (2.08 MB)

Non-trainable params: 0 (0.00 B)

matches hand count 545,098? True

```

In [11]: U.set_seed(U.SEED)
with U.Timer() as t_keras:
    hk = keras_model.fit(Xtr_k, y_train, validation_data=(Xte_k, y_test),
                        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2)
    print(f"\ntotal training time: {t_keras.seconds:.1f}s")

```

Epoch 1/5

79/79 - 5s - 58ms/step - accuracy: 0.3464 - loss: 1.8190 - val_accuracy: 0.4655 - val_loss: 1.4890

Epoch 2/5

79/79 - 3s - 35ms/step - accuracy: 0.5062 - loss: 1.3708 - val_accuracy: 0.5130 - val_loss: 1.3491

Epoch 3/5

```

79/79 - 3s - 36ms/step - accuracy: 0.6036 - loss: 1.1250 - val_accuracy: 0.5390 - va
l_loss: 1.2910
Epoch 4/5
79/79 - 3s - 35ms/step - accuracy: 0.6838 - loss: 0.9327 - val_accuracy: 0.5450 - va
l_loss: 1.2738
Epoch 5/5
79/79 - 3s - 36ms/step - accuracy: 0.7396 - loss: 0.7790 - val_accuracy: 0.5450 - va
l_loss: 1.3610
total training time: 16.1s

```

```

In [12]: pred_keras = keras_model.predict(Xte_k, batch_size=256, verbose=0).argmax(axis=1)
m = U.evaluate(y_test, pred_keras, N_CLASSES)

hist_keras = {"loss": [float(v) for v in hk.history["loss"]],
              "val_acc": [float(v) for v in hk.history["val_accuracy"]]}

results.append(U.RunResult(
    framework="TensorFlow/Keras", dataset="CIFAR-10 (5k subset)", model="SmallCNN",
    n_params=int(keras_model.count_params()), epochs=EPOCHS,
    train_seconds=round(t_keras.seconds, 2),
    train_loss=round(hist_keras["loss"][-1], 4),
    history=hist_keras, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")

```

accuracy 0.5450 macro-F1 0.5380

5. Implementation C — PyTorch

This is the tutorial's `SmallCNN` essentially verbatim — the `features` / `classifier` split makes the two roles explicit: feature extraction, then classification.

```

In [13]: U.set_seed(U.SEED)

class SmallCNN(nn.Module):
    def __init__(self, in_ch, n_classes):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(in_ch, CH1, K, padding=PAD),
            nn.ReLU(),
            nn.MaxPool2d(2),
            nn.Conv2d(CH1, CH2, K, padding=PAD),
            nn.ReLU(),
            nn.MaxPool2d(2),
        )
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(FLAT, HIDDEN),
            nn.ReLU(),
            nn.Linear(HIDDEN, n_classes),
        )

    def forward(self, x):

```

```

        x = self.features(x)
        x = self.classifier(x)
        return x

torch_model = SmallCNN(C, N_CLASSES).to(DEVICE)
n_torch = sum(p.numel() for p in torch_model.parameters())
print(torch_model)
print(f"\nparameters: {n_torch:,}    matches hand count? {n_torch == EXPECTED_PARAMS}")

SmallCNN(
  (features): Sequential(
    (0): Conv2d(3, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): ReLU()
    (2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (3): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (4): ReLU()
    (5): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=4096, out_features=128, bias=True)
    (2): ReLU()
    (3): Linear(in_features=128, out_features=10, bias=True)
  )
)
parameters: 545,098    matches hand count? True

```

```

In [14]: def train_torch(model, Xtr, ytr, Xte, yte, epochs=EPOCHS, batch_size=BATCH_SIZE, lr=
# Explicit loop: zero grad -> forward -> loss -> backward -> update.
loader = DataLoader(TensorDataset(torch.tensor(Xtr), torch.tensor(ytr)),
                    batch_size=batch_size, shuffle=True)
Xte_t = torch.tensor(Xte).to(DEVICE)
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=lr)
history = {"loss": [], "val_acc": []}

with U.Timer() as t:
    for epoch in range(1, epochs + 1):
        model.train()
        running, nb = 0.0, 0
        for xb, yb in loader:
            xb, yb = xb.to(DEVICE), yb.to(DEVICE)
            optimizer.zero_grad()
            loss = criterion(model(xb), yb)
            loss.backward()
            optimizer.step()
            running += loss.item(); nb += 1

        model.eval()
        with torch.no_grad():
            preds = [model(Xte_t[i:i + 1024]).argmax(1).cpu().numpy()
                      for i in range(0, len(Xte_t), 1024)]
            acc = float((np.concatenate(preds) == yte).mean())

        history["loss"].append(running / nb)
        history["val_acc"].append(acc)

```

```

        if log:
            print(f"epoch {epoch}/{epochs}  loss {running/nb:.4f}  test_acc {acc:.4f}")
    return history, t.seconds

```

```

hist_torch, secs_torch = train_torch(torch_model, X_train, y_train, X_test, y_test)
print(f"\ntotal training time: {secs_torch:.1f}s")

```

```

epoch 1/5  loss 1.8927  test_acc 0.4125
epoch 2/5  loss 1.4790  test_acc 0.4800
epoch 3/5  loss 1.2810  test_acc 0.5140
epoch 4/5  loss 1.1476  test_acc 0.5160
epoch 5/5  loss 0.9803  test_acc 0.5290
total training time: 3.1s

```

```

In [15]: torch_model.eval()
        with torch.no_grad():
            pred_torch = torch_model(torch.tensor(X_test).to(DEVICE)).argmax(1).cpu().numpy()
            m = U.evaluate(y_test, pred_torch, N_CLASSES)

        results.append(U.RunResult(
            framework="PyTorch", dataset="CIFAR-10 (5k subset)", model="SmallCNN",
            n_params=n_torch, epochs=EPOCHS, train_seconds=round(secs_torch, 2),
            train_loss=round(hist_torch["loss"][-1], 4), history=hist_torch, **m,
        ))
        print(f"accuracy {m['test_accuracy']:.4f}  macro-F1 {m['f1_macro']:.4f}")

```

```

accuracy 0.5290  macro-F1 0.5238

```

6. Comparison on the shared subset

```

In [16]: table = U.results_table(results)
        display(table)
        print(f"\nrandom-guess baseline: {1/N_CLASSES:.4f}")

```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_acc
0	Scratch (NumPy)	CIFAR-10 (5k subset)	SmallCNN	545098	5	1495.13	1.0495	
1	TensorFlow/Keras	CIFAR-10 (5k subset)	SmallCNN	545098	5	16.09	0.7790	
2	PyTorch	CIFAR-10 (5k subset)	SmallCNN	545098	5	3.10	0.9803	

```

random-guess baseline: 0.1000

```

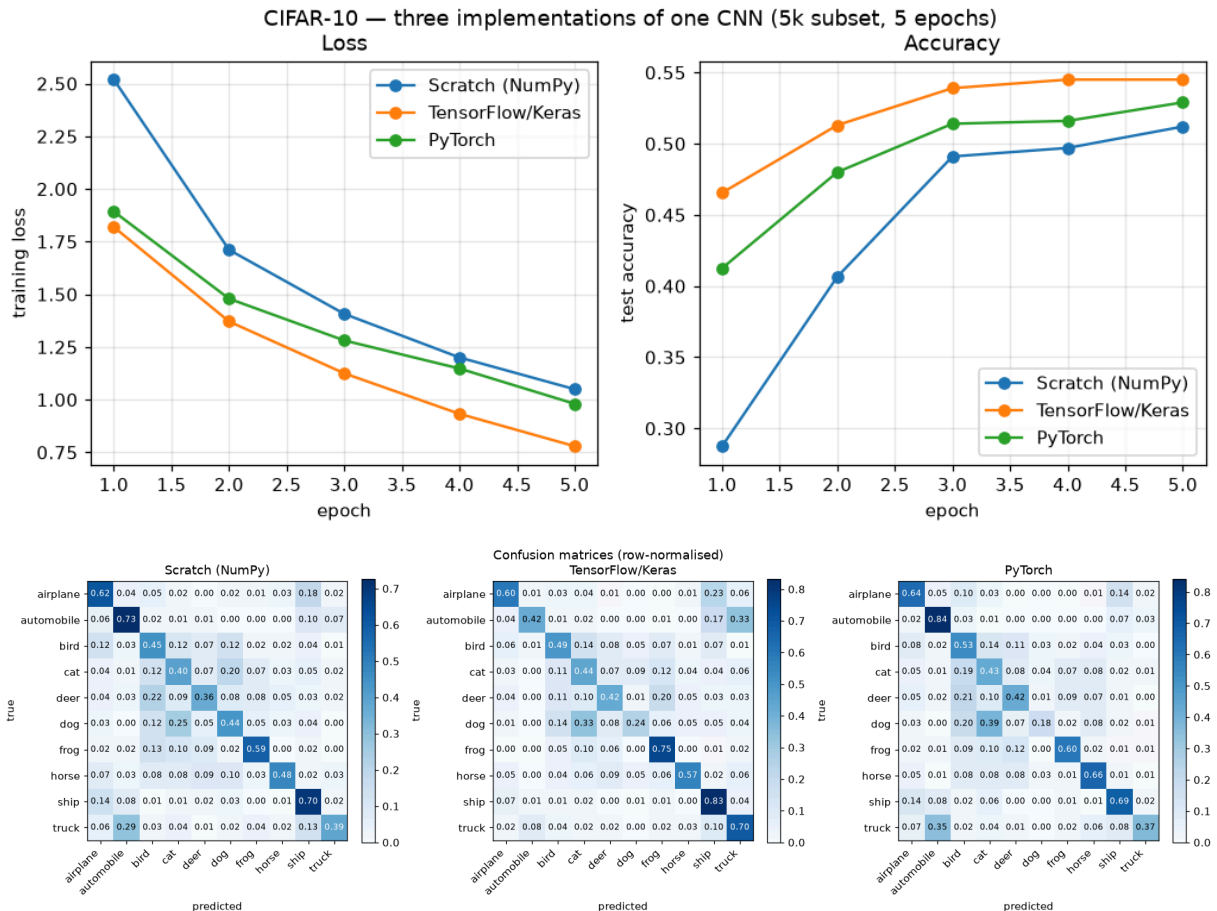
```

In [17]: U.plot_history({r.framework: r.history for r in results},
                    title="CIFAR-10 - three implementations of one CNN (5k subset, 5 epo

```

```
plt.show()

fig, axes = plt.subplots(1, 3, figsize=(16, 4.8))
for ax, r in zip(axes, results):
    U.plot_confusion(r.confusion, CLASSES, title=r.framework, ax=ax)
fig.suptitle("Confusion matrices (row-normalised)")
fig.tight_layout()
plt.show()
```



```
In [18]: cm = np.array(results[-1].confusion)
np.fill_diagonal(cm, 0)
pairs = np.vstack(np.unravel_index(np.argsort(cm.ravel())[:-1], cm.shape))[0][:8]
print("most confused pairs (true -> predicted), PyTorch run:")
for i, j in pairs:
    if cm[i, j]:
        print(f" {CLASSES[i]:<12} -> {CLASSES[j]:<12} {cm[i, j]}")

wrong = np.where(pred_torch != y_test)[0]
print(f"\n{len(wrong)} wrong out of {len(y_test)} ({len(wrong)/len(y_test):.1%}")
if len(wrong):
    sel = wrong[:20]
    U.plot_samples(X_test[sel], y_test[sel], CLASSES, n=len(sel),
                  preds=pred_torch[sel], title="PyTorch SmallCNN - misclassified i
plt.show()
```

most confused pairs (true -> predicted), PyTorch run:

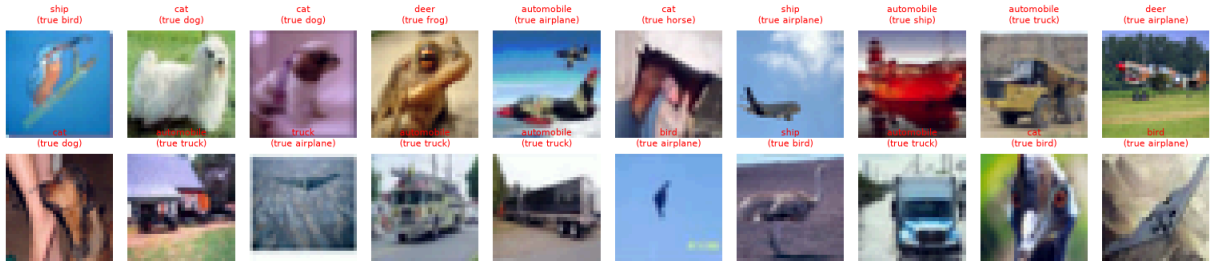
```

dog      -> cat      85
truck    -> automobile 69
deer     -> bird     44
dog      -> bird     43
cat      -> bird     42
airplane -> ship     29
ship     -> airplane 27
frog     -> deer     25

```

942 wrong out of 2000 (47.1%)

PyTorch SmallCNN — misclassified images



7. Full-data reference runs

Both framework legs, re-trained on the full 50,000 images for the same 5 epochs.

```

In [19]: Xtr_f, ytr_f, Xte_f, yte_f, _ = U.load_cifar10(verbose=True)
full_results = []

U.set_seed(U.SEED)
torch_full = SmallCNN(C, N_CLASSES).to(DEVICE)
hist_t_full, secs_t_full = train_torch(torch_full, Xtr_f, ytr_f, Xte_f, yte_f)

torch_full.eval()
with torch.no_grad():
    Xte_ft = torch.tensor(Xte_f).to(DEVICE)
    pred_full_torch = np.concatenate(
        [torch_full(Xte_ft[i:i + 1024]).argmax(1).cpu().numpy()
         for i in range(0, len(Xte_ft), 1024)])

m = U.evaluate(yte_f, pred_full_torch, N_CLASSES)
full_results.append(U.RunResult(
    framework="PyTorch", dataset="CIFAR-10 (full 50k)", model="SmallCNN",
    n_params=n_torch, epochs=EPOCHS, train_seconds=round(secs_t_full, 2),
    train_loss=round(hist_t_full["loss"][-1], 4), history=hist_t_full, **m,
))
print(f"\nPyTorch full-data: accuracy {m['test_accuracy']:.4f} macro-F1 {m['f1_mac

```

CIFAR-10: train (50000, 3, 32, 32) / test (10000, 3, 32, 32), 10 classes

epoch 1/5 loss 1.3002 test_acc 0.6348

epoch 2/5 loss 0.9251 test_acc 0.6935

epoch 3/5 loss 0.7666 test_acc 0.6979

epoch 4/5 loss 0.6490 test_acc 0.7113

epoch 5/5 loss 0.5490 test_acc 0.7184

PyTorch full-data: accuracy 0.7184 macro-F1 0.7167

```

In [20]: U.set_seed(U.SEED)
keras_full = build_keras_cnn("cifar_cnn_full")
with U.Timer() as t_kf:
    hkf = keras_full.fit(np.transpose(Xtr_f, (0, 2, 3, 1)), ytr_f,
                        validation_data=(np.transpose(Xte_f, (0, 2, 3, 1)), yte_f),
                        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2)

pred_full_keras = keras_full.predict(np.transpose(Xte_f, (0, 2, 3, 1)),
                                     batch_size=512, verbose=0).argmax(axis=1)
m = U.evaluate(yte_f, pred_full_keras, N_CLASSES)
full_results.append(U.RunResult(
    framework="TensorFlow/Keras", dataset="CIFAR-10 (full 50k)", model="SmallCNN",
    n_params=int(keras_full.count_params()), epochs=EPOCHS, train_seconds=round(t_kf, 4),
    train_loss=round(float(hkf.history["loss"][-1]), 4),
    history={"loss": [float(v) for v in hkf.history["loss"]],
            "val_acc": [float(v) for v in hkf.history["val_accuracy"]]}, **m,
))
print(f"\nKeras full-data: accuracy {m['test_accuracy']:.4f} macro-F1 {m['f1_macro']}")

```

Epoch 1/5

782/782 - 28s - 36ms/step - accuracy: 0.5581 - loss: 1.2410 - val_accuracy: 0.6507 - val_loss: 0.9999

Epoch 2/5

782/782 - 26s - 34ms/step - accuracy: 0.6930 - loss: 0.8850 - val_accuracy: 0.6923 - val_loss: 0.8886

Epoch 3/5

782/782 - 26s - 33ms/step - accuracy: 0.7466 - loss: 0.7274 - val_accuracy: 0.6974 - val_loss: 0.9091

Epoch 4/5

782/782 - 26s - 33ms/step - accuracy: 0.7924 - loss: 0.5996 - val_accuracy: 0.7005 - val_loss: 0.9604

Epoch 5/5

782/782 - 26s - 33ms/step - accuracy: 0.8293 - loss: 0.4958 - val_accuracy: 0.7011 - val_loss: 1.0141

Keras full-data: accuracy 0.7011 macro-F1 0.6972

```

In [21]: display(U.results_table(results + full_results))

sub_best = max(r.test_accuracy for r in results)
full_best = max(r.test_accuracy for r in full_results)
print(f"\nbest on 5k subset : {sub_best:.4f}")
print(f"best on full 50k : {full_best:.4f}")
print(f"cost of the subset: {full_best - sub_best:+.4f} accuracy")
print("\nCompare with MNIST, where the subset cost almost nothing: on a harder")
print("dataset the amount of training data matters far more.")

```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_acc
0	Scratch (NumPy)	CIFAR-10 (5k subset)	SmallCNN	545098	5	1495.13	1.0495	C
1	TensorFlow/Keras	CIFAR-10 (5k subset)	SmallCNN	545098	5	16.09	0.7790	C
2	PyTorch	CIFAR-10 (5k subset)	SmallCNN	545098	5	3.10	0.9803	C
3	PyTorch	CIFAR-10 (full 50k)	SmallCNN	545098	5	17.67	0.5490	C
4	TensorFlow/Keras	CIFAR-10 (full 50k)	SmallCNN	545098	5	133.12	0.4958	C

best on 5k subset : 0.5450

best on full 50k : 0.7184

cost of the subset: +0.1734 accuracy

Compare with MNIST, where the subset cost almost nothing: on a harder dataset the amount of training data matters far more.

8. Where each component lives

```
In [22]: component_table = pd.DataFrame([
    ["Data loading", "ass4_utils -> NumPy (N,C,H,W)", "same arrays, (N,H,W,C)",
    ["Preprocessing", "ass4_utils (shared)", "shared",
    ["Model definition", "S.Sequential([...])", "keras.Sequential([...])",
    ["Convolution", "S.Conv2D (im2col + matmul)", "keras.layers.Conv2D",
    ["Activation", "S.ReLU", "activation='relu'",
    ["Pooling", "S.MaxPool2D", "keras.layers.MaxPooling2D",
    ["Flatten", "S.Flatten", "keras.layers.Flatten",
    ["Dense layer", "S.Dense", "keras.layers.Dense",
    ["Loss", "S.SoftmaxCrossEntropy", "SparseCategoricalCrossent
    ["Gradient", "hand-derived backward + col2im", "automatic (GradientTape)",
    ["Optimizer", "S.Adam (written out)", "keras.optimizers.Adam",
    ["Training loop", "explicit for-loop", "model.fit()",
    ["Evaluation", "predict_classes + sklearn", "predict + sklearn",
    ["Hardware", "CPU (NumPy)", "CPU (no GPU TF on Windows
], columns=["Component", "Scratch", "TensorFlow/Keras", "PyTorch"])

display(component_table)

os.makedirs("results", exist_ok=True)
with open("results/03_cifar10.json", "w") as f:
    json.dump({"meta": dict(meta),
               "results": [r.__dict__ for r in results],
```

```
    "full_results": [r.__dict__ for r in full_results],
    "component_table": component_table.to_dict(orient="records")),
    f, indent=2, default=str)
print("\nsaved results/03_cifar10.json")
```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy (N,C,H,W)	same arrays, (N,H,W,C)	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module
3	Convolution	S.Conv2D (im2col + matmul)	keras.layers.Conv2D	nn.Conv2d
4	Activation	S.ReLU	activation='relu'	nn.ReLU
5	Pooling	S.MaxPool2D	keras.layers.MaxPooling2D	nn.MaxPool2d
6	Flatten	S.Flatten	keras.layers.Flatten	nn.Flatten
7	Dense layer	S.Dense	keras.layers.Dense	nn.Linear
8	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropyLoss
9	Gradient	hand-derived backward + col2im	automatic (GradientTape)	automatic (autograd)
10	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
11	Training loop	explicit for-loop	model.fit()	explicit for-loop
12	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn
13	Hardware	CPU (NumPy)	CPU (no GPU TF on Windows)	GPU (cuDNN)

saved results/03_cifar10.json

Saving the trained models

The JSON above records what the models *scored*. It does not record what they *are* — rerunning any of these numbers would mean retraining from scratch.

`U.save_model` writes each leg in its framework's native format, under `results/models/<notebook>/`, with a JSON sidecar naming the architecture and the score the weights actually produced:

leg	file	reloads alone?
Scratch (NumPy)	<code>.npz</code> , one array per layer parameter	no
TensorFlow/Keras	<code>.keras</code> , graph + weights	yes

leg	file	reloads alone?
PyTorch	.pt , state_dict	no

.pt and .npz hold weights only, so reloading them means rebuilding the architecture first and passing the fresh instance to `U.load_model(..., model=...)`. That is the same contract `variants.py` uses for `results/variant_cache/`.

```
In [23]: # Weights are the model; metrics only describe it. Persist all three Legs.
TAG = "03_cifar10"

by_fw = {r.framework: r for r in results}

for obj, fw, stem in [
    (scratch_model, "Scratch (NumPy)", "smallcnn_scratch"),
    (keras_model, "TensorFlow/Keras", "smallcnn_keras"),
    (torch_model, "PyTorch", "smallcnn_torch"),
]:
    r = by_fw[fw]
    U.save_model(obj, f"{stem}_subset", TAG, architecture=r.model, dataset=r.dataset,
                 epochs=r.epochs, test_accuracy=r.test_accuracy, f1_macro=r.f1_macro)

# The full-data reruns - scratch NumPy is not rerun at this size, so two legs.
by_fw_full = {r.framework: r for r in full_results}

for obj, fw, stem in [
    (keras_full, "TensorFlow/Keras", "smallcnn_keras"),
    (torch_full, "PyTorch", "smallcnn_torch"),
]:
    r = by_fw_full[fw]
    U.save_model(obj, f"{stem}_full", TAG, architecture=r.model, dataset=r.dataset,
                 epochs=r.epochs, test_accuracy=r.test_accuracy, f1_macro=r.f1_macro)

display(U.list_models(TAG)[["name", "framework", "n_params", "dataset", "test_accu"])
```

saved results\models\03_cifar10\smallcnn_scratch_subset.npz (545,098 params, 1979 K B)

saved results\models\03_cifar10\smallcnn_keras_subset.keras (545,098 params, 6425 K B)

saved results\models\03_cifar10\smallcnn_torch_subset.pt (545,098 params, 2133 KB)

saved results\models\03_cifar10\smallcnn_keras_full.keras (545,098 params, 6425 KB)

saved results\models\03_cifar10\smallcnn_torch_full.pt (545,098 params, 2133 KB)

	name	framework	n_params	dataset	test_accuracy
0	smallcnn_keras_full	keras	545098	CIFAR-10 (full 50k)	0.7011
1	smallcnn_keras_subset	keras	545098	CIFAR-10 (5k subset)	0.5450
2	smallcnn_scratch_subset	scratch	545098	CIFAR-10 (5k subset)	0.5120
3	smallcnn_torch_full	torch	545098	CIFAR-10 (full 50k)	0.7184
4	smallcnn_torch_subset	torch	545098	CIFAR-10 (5k subset)	0.5290

9. What this dataset shows

The three implementations still agree. Same 545,098 parameters, same shapes, comparable accuracy. Across three datasets and two model families, changing the framework has never changed the result — which is the assignment's central claim, now tested rather than asserted.

But the model itself is clearly inadequate here. Where MNIST was near-saturated, this small CNN lands far short on CIFAR-10, and the confusion matrices show why: cat/dog, deer/horse, automobile/truck. These are classes that need texture and part structure, not just edges. Five epochs and a two-convolution stack cannot supply that.

And the parameters are in the wrong place. 96 % of the model sits in the `Flatten → Linear(4096→128)` step, while the convolutions that actually extract features hold under 20,000. Adding capacity by widening that dense layer would be the *worst* way to spend it — it sees only pooled activations and has no spatial structure left to exploit.

That diagnosis is the bridge to the last notebook. Tutorial §45 asks for exactly the experiment this situation calls for:

```
$$M_1: Conv{+}ReLU{+}Pool \qqquad M_2: M_1 {+} BatchNorm \qqquad M_3: M_2 {+} Residual
\qqquad M_4: M_3 {+} Attention$$
```

Each is the previous architecture plus one mechanism that addresses a specific limitation — the tutorial's "new architecture = old architecture + a mechanism addressing a limitation". CIFAR-10 is hard enough for those differences to show up in the numbers.

Next: `04_compare.ipynb`.